Who Pays for Oil Shocks? Fuel Price Pass-Through to the Inflation of Poor Households across Philippine Regions

Capstone Paper Presentation (SA3)

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1. Why this question matters now

The 2026 oil shock pushed inflation above target, and the poor's inflation even higher.

The shock

- **Pump prices:** the fuel index was 76% above a year earlier in April 2026
- **Headline inflation:** 2.4% in February → 7.2% in April → 6.2% in August
- **Bottom 30% of households:** 8.5% in April and 8.2% in August
- **BSP target:** 2–4%

The policy response

- **RA 12316:** lets the President suspend fuel excise taxes when Dubai crude ≥ USD 80/barrel
- **Excise suspended** on LPG and kerosene, not on gasoline and diesel
- **Fuel subsidies** for public utility vehicle drivers
- **BSP** raised its policy rate three times, to 5.00%

Sources: PSA CPI (2018 = 100); RA 12316; BSP. Inflation is year on year.

2. Research gap and questions

Cross-country evidence says fuel shocks hurt the rich more; is that true in the Philippines?

The gap

- **Kpodar and Liu (2022), 122 countries:** fuel price shocks are *progressive*: prices rise more for richer households
- **But Filipino poor households** spend over half their budget on food and ride public transport
- **PSA publishes a bottom-30% CPI** by region, so we can test this directly

Three objectives

- **Objective 1 (descriptive):** trends in fuel prices and both inflation rates, 2001–2026
- **Objective 2 (estimation):** pass-through (RQ1), the inflation gap (RQ2), categories (RQ3), regions (RQ4)
- **Objective 3 (policy):** the 2026 shock, its peso cost and the policy response

Research question: When fuel prices rise, do poor Filipino households face larger price increases than the average household, and does it depend on where they live?

3. Key terms and notation

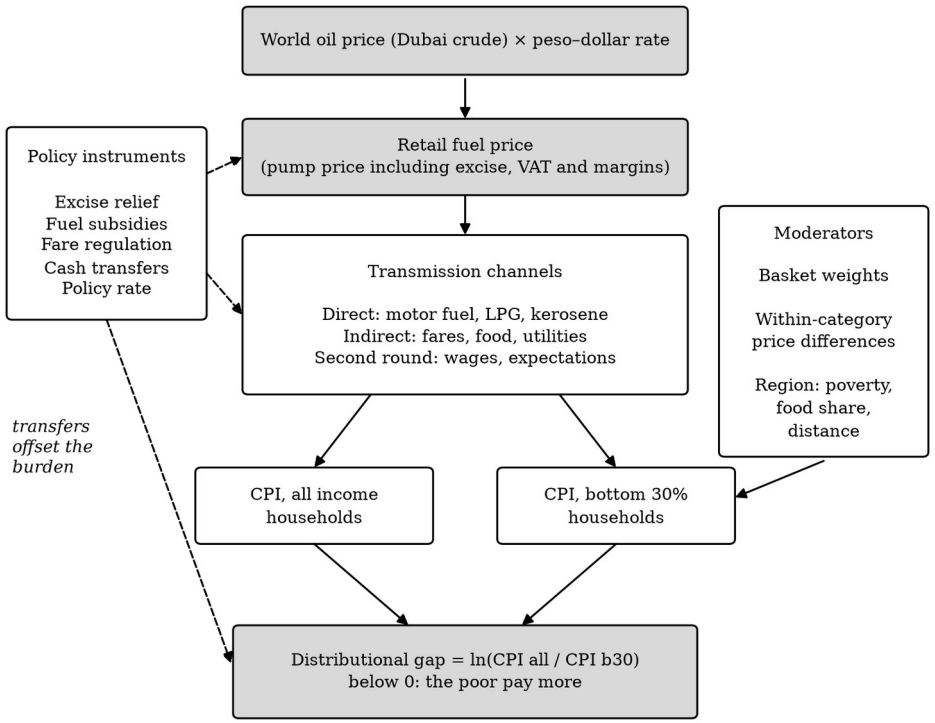
Each term keeps the same meaning on every slide.

Term	Meaning and formula
Fuel price shock	Monthly change in PSA's retail fuel index 07.2.2: $f_t = 100 \cdot \Delta \ln F_t$
Inflation (group g)	$\pi^g_t = 100 \cdot \Delta \ln P^g_t$; A = all income households, B = bottom 30%
Pass-through	β_h: change in inflation h months later per 1% fuel rise; cumulative $B_h = \beta_0 + \ldots + \beta_h$
Inflation gap	$D_t = 100 \cdot \ln(P^A_t / P^B_t)$; a fall means prices rose more for the poor (*regressive*)
Basket weight	w^g_k: share of group g's 2018 budget spent on category k
Region groups	Our own dummies G_r: high poverty, high food share, Mindanao, island regions

Both CPIs are official PSA indices; the gap and the region groups are our constructions.

4. Theoretical framework

A group's pass-through equals the fuel content of what it buys.



Solid arrows: transmission of the shock. Dashed arrows: points where policy can intervene.

Solid: transmission; dashed: where policy acts.

- **Cost-push pricing** (Leontief input–output): each good's price rises with its fuel cost share, $\pi_k \approx \varphi_k f$
- **Cost-of-living index:** $\pi^g \approx \Sigma_k w^g_k \varphi_k f$, so pass-through = fuel intensity of the basket
- **Gap = weights effect + within-category effect** (the poor buy different items in each category)
- **Oil-specific channels:** direct, indirect and second-round (Kilian 2008; Blanchard and Galí 2010)
- **Welfare cost** ≈ budget-share-weighted price rise (Deaton 1989)

5. Data and empirical strategy

Local projections on monthly PSA data, 2001–August 2026, national and 17 regions.

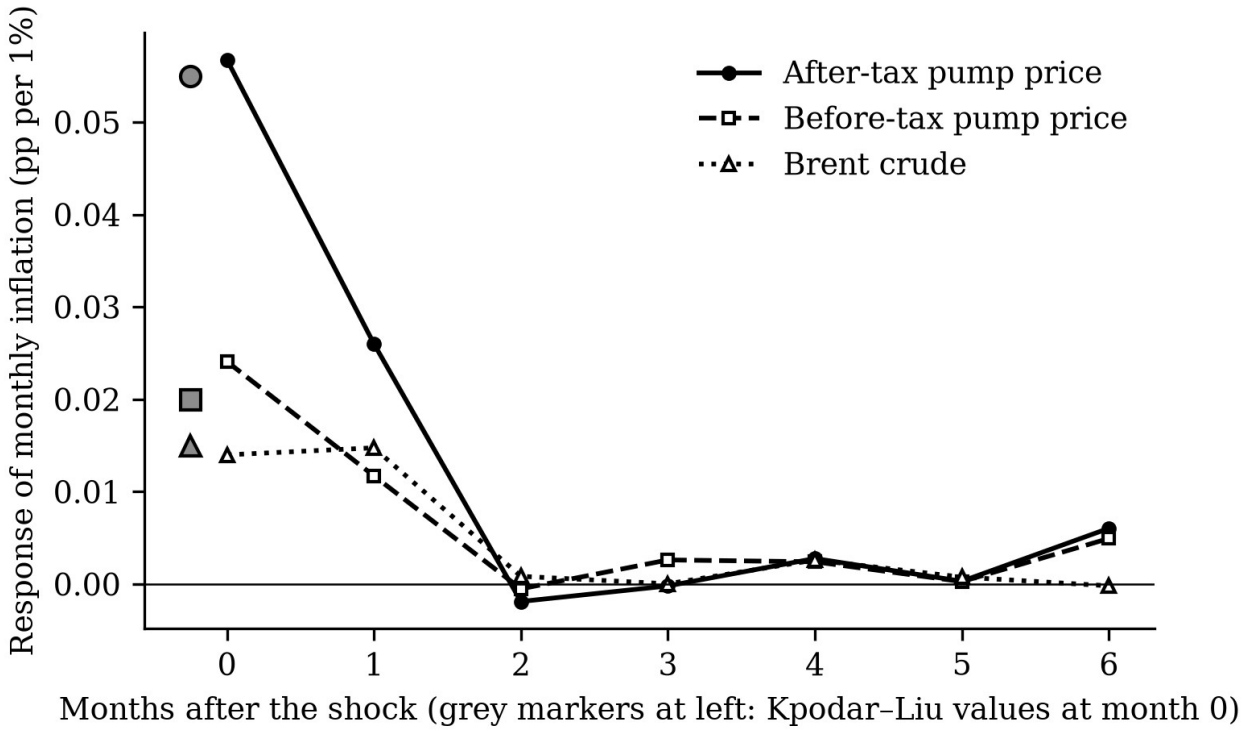
Variable	Source	Coverage
All-income CPI and 13 categories	PSA OpenSTAT	1994–Aug 2026, national and 17 regions
Bottom-30% CPI	PSA OpenSTAT	2000–Aug 2026, national and 17 regions
Retail fuel index 07.2.2 (shock)	PSA OpenSTAT	1994–Aug 2026, national and 17 regions
Crude oil, pump price, FX, policy rate, food	World Bank, BSP, BIS, FAO	Robustness and controls

Model (Jordà 2005; Kpodar and Liu 2022), for each h = 0…12: $\pi_{t+h} = \alpha_h + \beta_h f_t$ + 12 lags + leads + trend + ε

Errors: Newey–West (national), region fixed effects and Driscoll–Kraay (panel); 90% bands

6. Validation before any Philippine estimate

Our code reproduces the reference study's European results under rules set in advance.

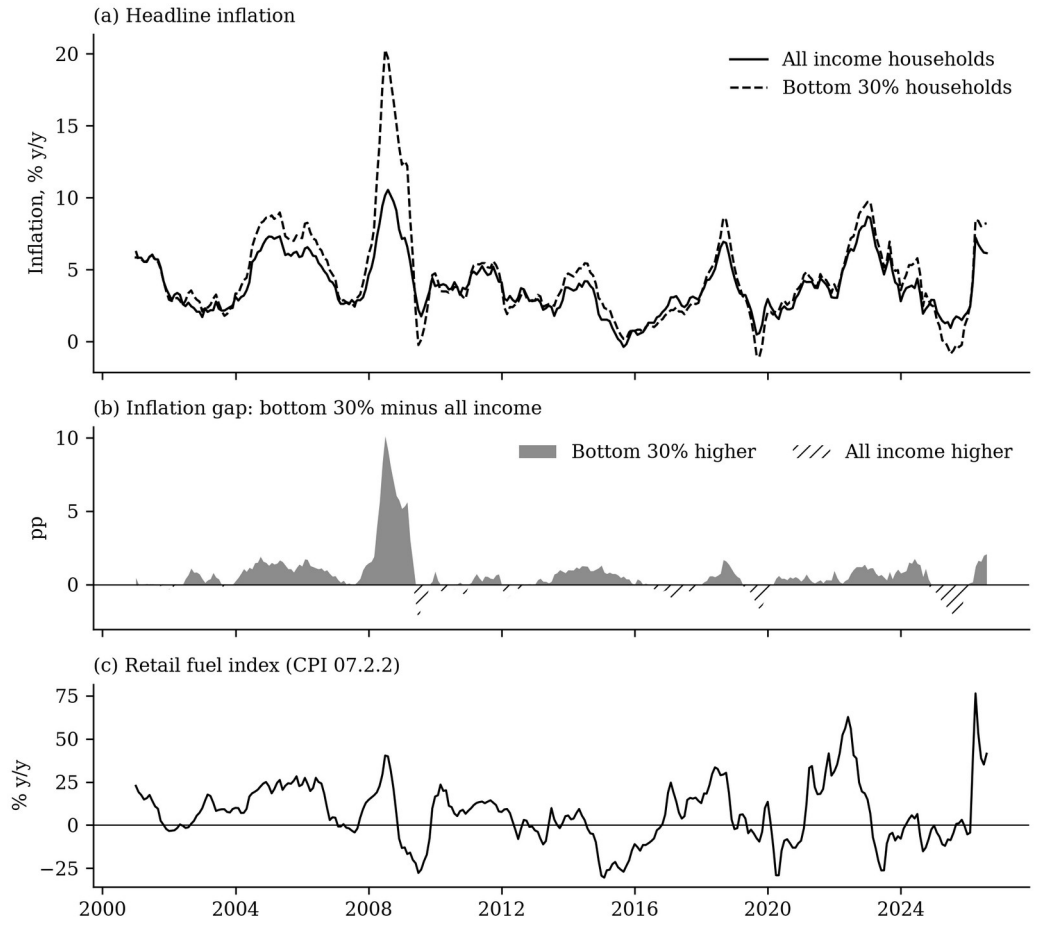


Shock (EU, impact)	Ours	Paper
Pump price after tax	0.057	0.055
Pump price before tax	0.024	0.020
Brent crude in euro	0.014	0.015

Response of monthly HICP inflation to a 1% fuel price rise, 2005–Jun 2019. A simulation test also passed.

7. Objective 1: the poor's inflation has been higher

Bottom-30% prices rose 13.3 log points more than all-income prices since 2001.

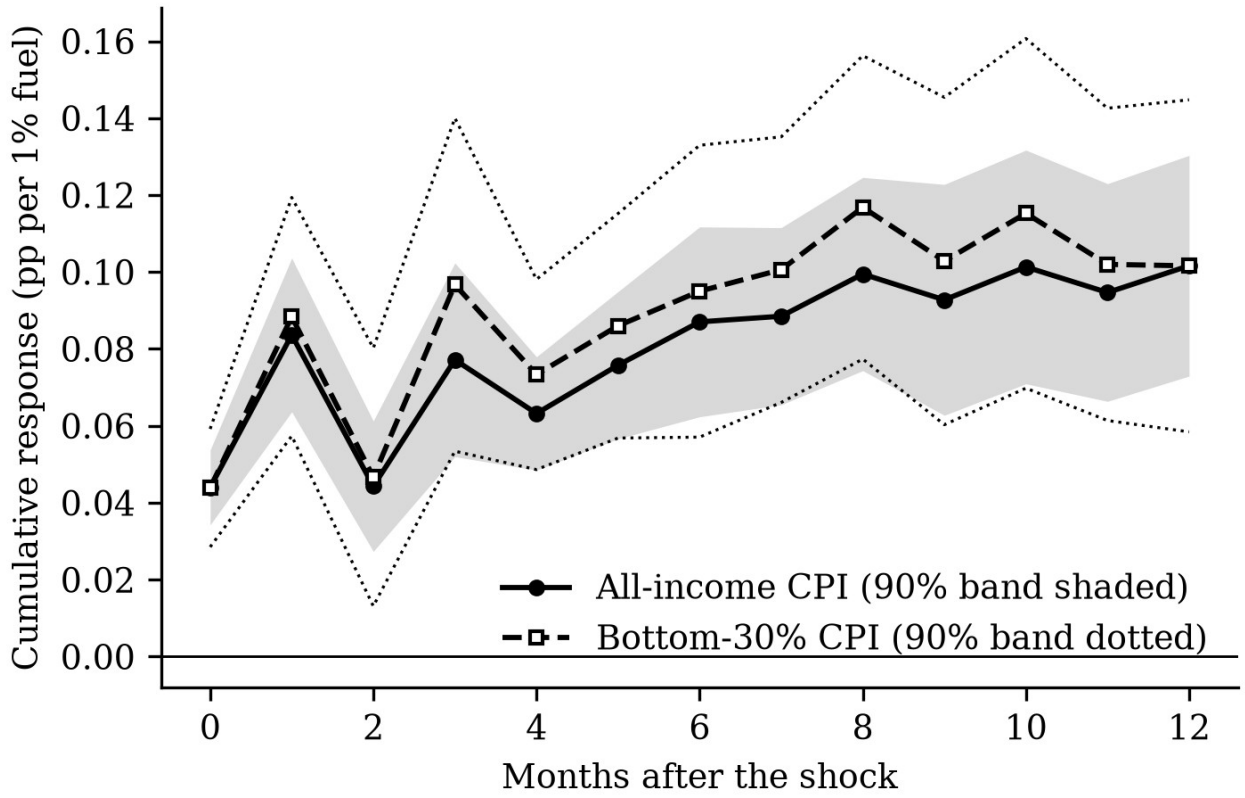


- **Average inflation, 2001–2026:** 4.33% (bottom 30%) vs 3.77% (all)
- **Higher for the poor** in 18 of 25 years
- **Fuel-shock episodes:** poor's inflation higher in 6 of 9
- **Peak gap in 2008** (rice crisis): 13.9% vs 8.2%

Source: PSA CPI; authors' calculations. Year-on-year rates.

8. RQ1: fuel prices pass through quickly and keep building

A 10% fuel price rise raises the price level by about 1.0% after a year.

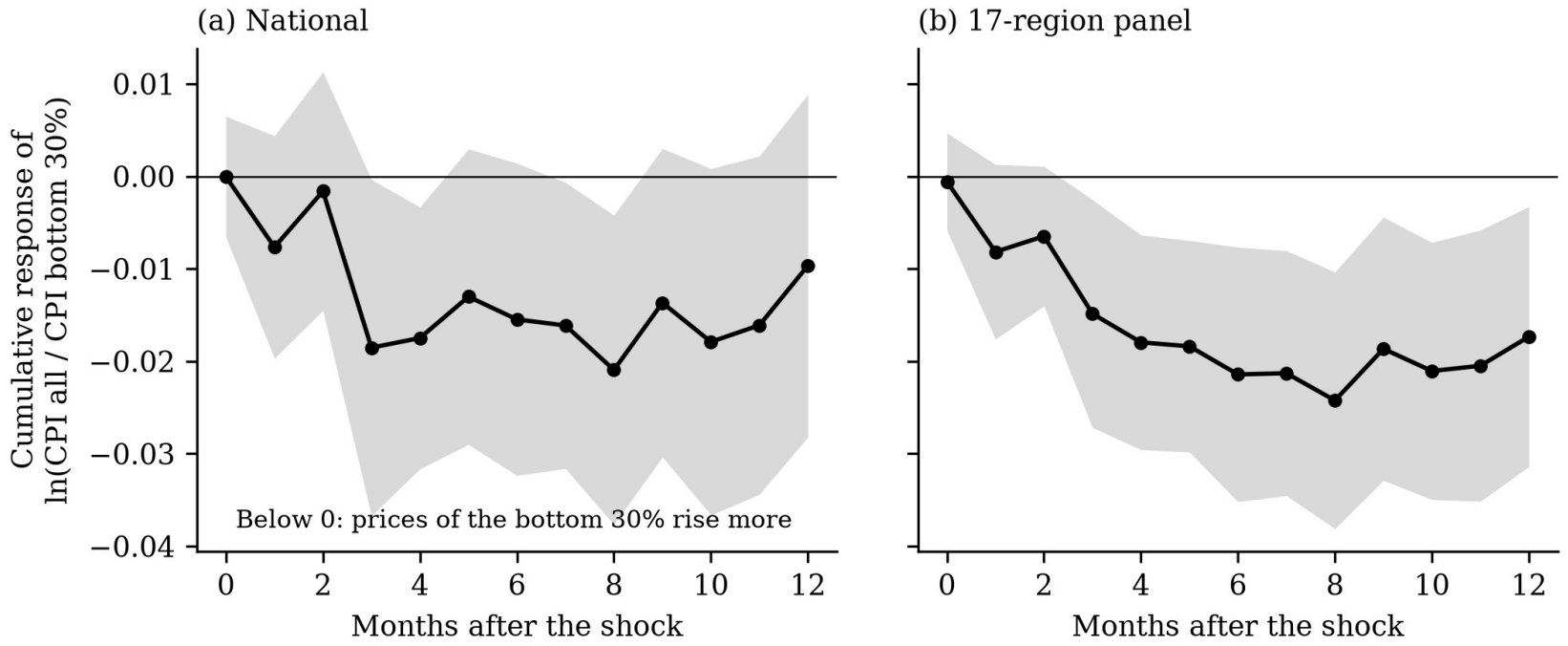


- **Impact:** 0.044 pp per 1% fuel rise (s.e. 0.006)
- **After 12 months:** 0.102 (all) and 0.102 (bottom 30%)
- **Larger than the cross-country** developing-economy impact (0.018)
- **Crude oil understates it:** only about 40% of the retail estimate

Cumulative responses, 90% bands. National, Newey–West standard errors.

9. RQ2: the poor pay more, not less

The gap falls after a fuel shock: the effect is regressive, the opposite of the cross-country finding.

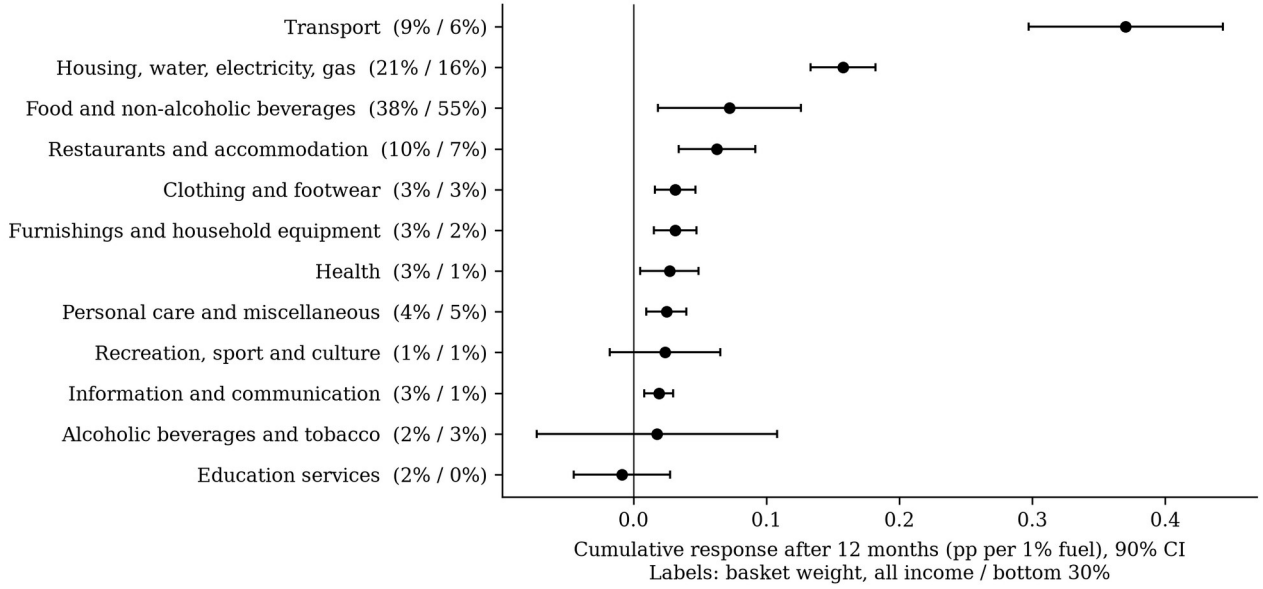


National: −0.0155 after 6 months (s.e. 0.0103), significant at h = 3, 4, 7, 8 · **17-region panel:** −0.0214 (s.e. 0.0084), significant

Meaning: a 10% fuel rise leaves the poor's price level about 0.2% higher than the average household's

10. RQ3: which prices carry the shock?

Transport, utilities and food carry the shock; budget shares alone would predict the opposite gap.

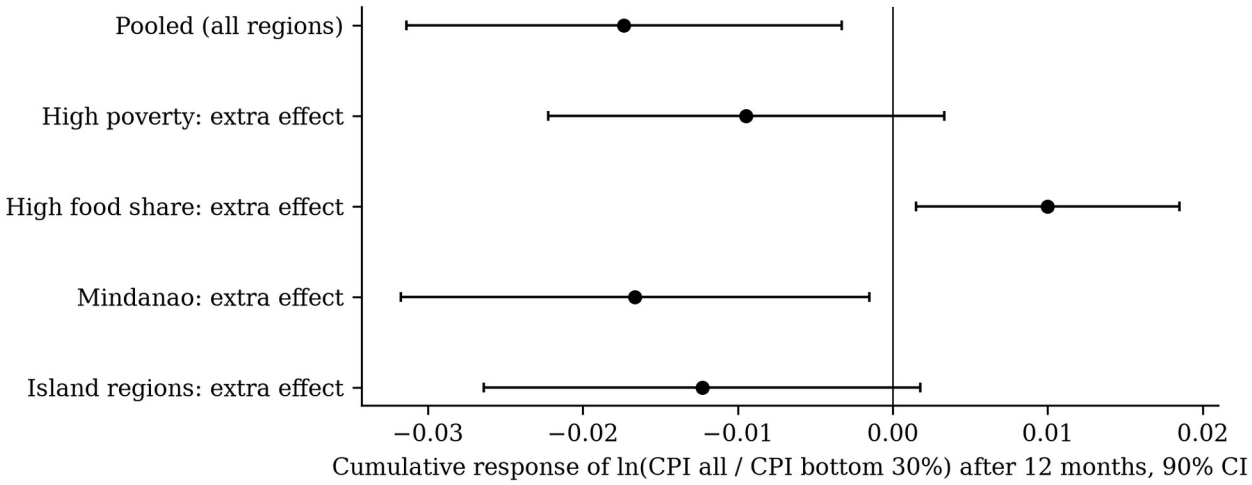


- **Largest responses:** transport 0.370, housing and utilities 0.158, food 0.072
- **10 of 13** categories respond significantly
- **Weights alone** predict a *progressive* gap (+0.0101)
- **Within categories, responses differ:** utilities 0.182 (poor) vs 0.142; transport 0.196 vs 0.300

Cumulative response after 12 months, all-income CPI divisions, 90% CI. Within-category comparison is exploratory (2013–2026).

11. RQ4: do regions differ?

The gap tends to be wider in Mindanao and island regions, but the evidence is weak.

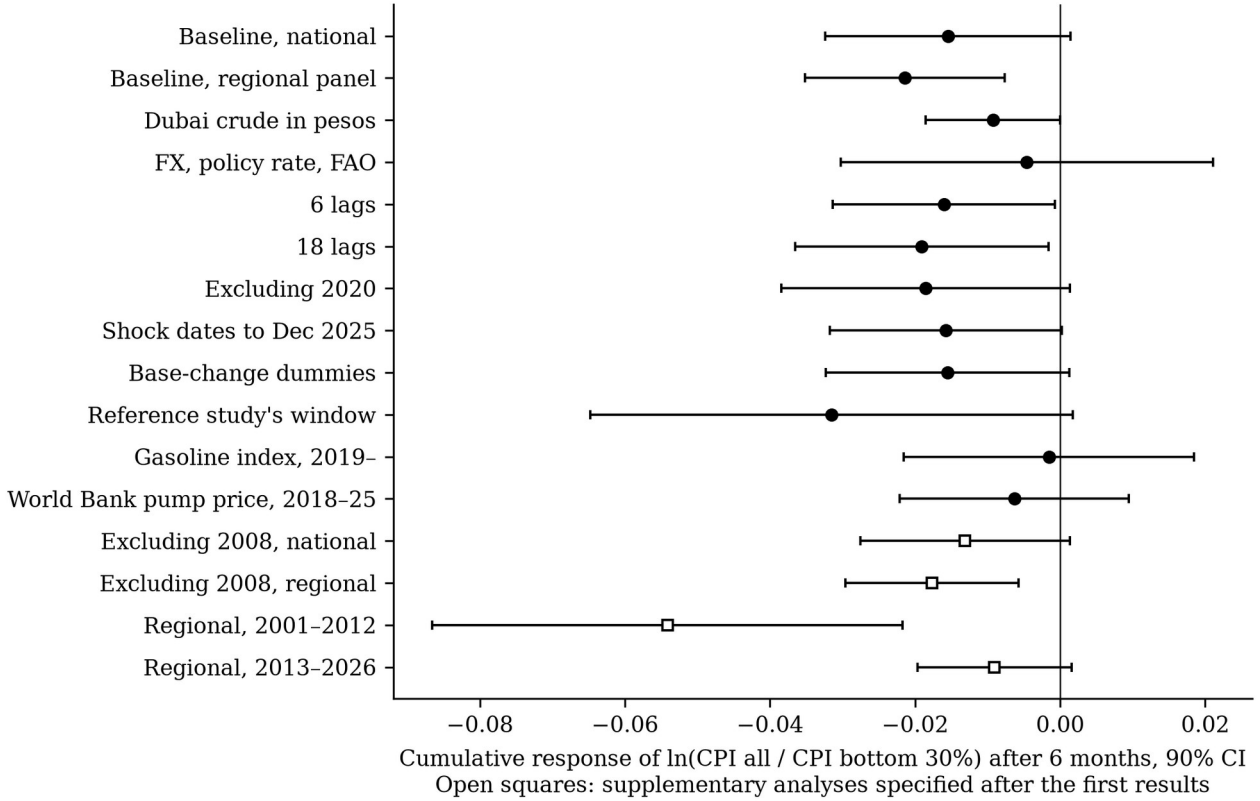


- **Mindanao:** −0.0166 extra (s.e. 0.0092)
- **Island regions:** −0.0123 (s.e. 0.0086)
- **High poverty:** −0.0095 (s.e. 0.0078)
- **High food share:** 0.0100 (s.e. 0.0052)
- **None survives** a correction for testing four groups

Additional cumulative effect on the gap after 12 months; 17-region panel, Driscoll–Kraay errors.

12. How robust is the regressive result?

The sign holds in almost every check, but the effect is concentrated in 2001–2012.

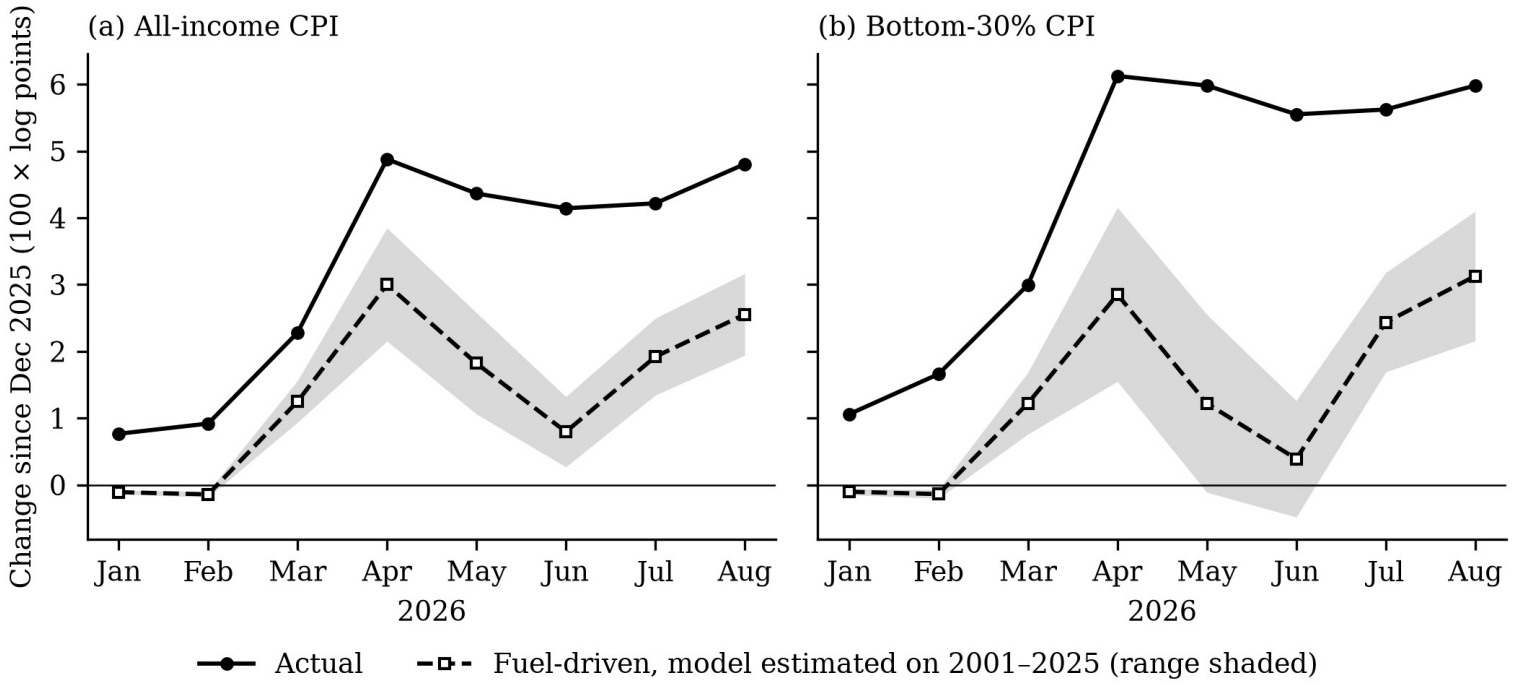


- **Leave one region out:** −0.0227 to −0.0203
- **2001–2012:** −0.0541; **2013–2026:** −0.0091
- **Near zero** with macro controls (−0.0046), the gasoline index and the World Bank pump price
- **Read with care:** regressive in sign, strongest in the 2000s

Cumulative gap response after 6 months, 90% CI. Open squares: supplementary analyses.

13. Objective 3: what explains the 2026 surge?

A model estimated only on data to 2025 explains about half of the 2026 price rise.



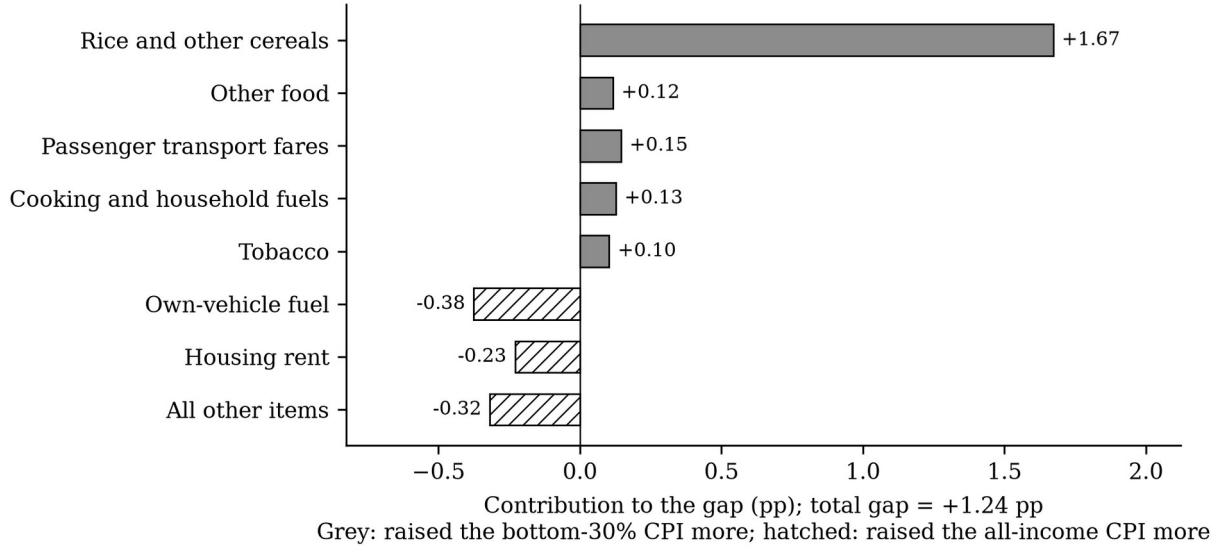
Dec 2025–Aug 2026: fuel-driven 2.55 of 4.81 log points (all), 3.13 of 5.98 (bottom 30%) · **Prices did not fall** when fuel eased in May–June

14. The cost to poor families, and why it was higher

Fuel added about PHP 467 a month to a poverty-line family's costs; rice drove the extra gap.

Peso cost by August 2026

- **National:** PHP 467 a month for a family of five at the poverty line
- **By region:** PHP 390 (XII SOCCSKSARGEN) to 633 (National Capital Region)
- **All price increases:** about PHP 907 a month
- **Highest actual rise:** XI Davao, BARMM, VI Western Visayas, XIII Caraga and IX Zamboanga Peninsula



Contributions to the +1.24 pp gap: cereals +1.67.

15. Policy implications

Transfers sized to the poor's own price index are better targeted than fuel tax cuts.

Instrument (2026)	What the evidence says
Excise relief, gasoline and diesel	Mainly helps the better-off: motor fuel is 2.39% of the average basket, 1.59% of the poor's
Excise relief, LPG and kerosene	Distribution-neutral for LPG (same weight and response); kerosene matters more for the poor but is small
Fuel subsidies for drivers	Protect poor commuters only if tied to fare restraint
Policy rate increases	Appropriate: pass-through builds for at least a year
Targeted cash transfers	Best targeted; size with the bottom-30% CPI (PHP 467–907 a month)
Rice price and supply measures	Needed: cereals explain +1.67 pp of the 2026 gap

16. Limitations and reproducibility

The results are descriptive of transmission, not a clean causal test, and every number can be rebuilt.

Limitations

- **Retail prices may respond** to local conditions; an identified oil supply shock is the next step
- **Bottom 30% vs all**, not poorest vs richest
- **2026 analysis** covers eight months; rice and trade policy cannot be separated from fuel

Data and reproducibility

- **All data public:** PSA OpenSTAT, World Bank, BSP, BIS, FAO, Eurostat
- **Data appendix** in the paper links every figure and table to its source and script
- **One command** (./run_all.sh) rebuilds every number
- **Pre-registered** estimation plan and replication test

TAKEAWAY

In the Philippines, the poor pay more when fuel prices rise.

- **Pass-through is fast and lasting:** a 10% fuel rise lifts prices about 1.0% within a year
- **Regressive, not progressive:** the poor's prices rise more, through food, utilities and fares
- **2026:** fuel explains about half of the surge; rice drove the poor's extra inflation
- **Policy:** protect the poor with transfers sized to their price index, not broad fuel tax cuts

Thank you

Questions and discussion